

cycle. Then the whole development process takes place again, with the four stages, to improve the software and then it is released again, with another version number. Once again the customer's feedback is taken, and if any improvements are required then the development cycle goes into the iteration stage again and the cycle keeps continuing until the software reaches maturity (a stable software). This type of development cycle could take an indefinite amount of time and causes delays in software development. That's why using DevOps can make the development cycle quicker and more effective.

With DevOps, the development team communicates with the operations team and does not deal with the customers. In DevOps, only small chunks of code are continuously coded and tested, which prevents any bottlenecks or delays in delivering the code. DevOps provides continuous delivery and integration of the code. This is why it is better than agile software development.

As DevOps has many benefits over agile software development, many software companies are resorting to the DevOps culture. Because it provides continuous delivery and integration, DevOps can help software products

and services to reach maturity in a very short period of time. DevOps is mainly suitable for the development of cloud computing products and services, because it needs collaboration between the development team and the IT operations team. As there is a growing demand for cloud computing products and services, DevOps is the right choice for most software companies and has a great future. **END** 

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Module management in Web applications using Gulp

The following is the hierarchy used for the files and folders while working with Gulp streaming applications.

src: Path of the pre-processed HTML code files:

Scripts: Assorted pre-processed scripts
Styles: Multiple Stylesheet Source Code in CSS format

Build: This folder is automatically created with the dynamic generation of production files:

Images: Compressed image files
Scripts: Script file with minified codes
Styles: CSS script with minified codes

gulpfile.js: Configuration file of Gulp in JavaScript for defining the tasks.

Task management and automation in Gulp

Gulp integrates the modular approach with task management so that the software development and related phases can be programmed with accuracy. You need to create a Gulp task with the following structure:

```
gulp.task('name of the task', function()
{
    // Operations
});
```

...where the *name of the task* refers to the title of the task. All the operations or tasks are defined in *function()*. The 'gulp.task' is used to register the function for the execution of the operations.

Here is an example:

```
gulp.task('mytask', function() {
    var SourcePath = 'src/myimages/**/*',
    DestinationPath = 'build/myimages';
    gulp.src(SourcePath)
    .pipe(changed(DestinationPath))
    .pipe(mytask())
    .pipe(gulp.dest(DestinationPath));
});
```

The execution or running of the Gulp file is done using the following instruction:

\$ gulp mytask

Today, large scale software projects have diversified teams of programmers, testing engineers as well as other team members. Hence, there is a need to automate and control tasks effectively so that companies can get fast delivery of the product, and to provide an integrated development environment with testing using the different tools of DevOps. The main requirement for the DevOps approach towards software development is that it should be cloud-centric so that simultaneous operations can be performed with the help of DevOps based tools and technologies. It is very difficult for small companies to follow DevOps because they can't afford cloud costs. From the point of view of a career, DevOps is in huge demand in the software development job market, with a very good salary package on offer. **END** 

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